CODEALPHA

PHARMACEUTICAL RESEARCH INTERNSHIP

Literature Review on Drug Development: Tirzepatide

A Review of Drug Discovery, Clinical Development, Therapeutic Applications and Market Impact

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Selected Drug	Tirzepatide (Mounjaro / Zepbound)
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Abstract

Tirzepatide is a novel once-weekly injectable peptide that activates both glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptors. Developed by Eli Lilly and Company as LY3298176, it represents a dual-incretin approach to metabolic disease. The U.S. Food and Drug Administration (FDA) approved tirzepatide under the brand name Mounjaro on 13 May 2022 to improve glycemic control in adults with type 2 diabetes mellitus (T2DM), and subsequently approved the same active ingredient as Zepbound for chronic weight management in adults with obesity or overweight with at least one weight-related condition. This review summarizes the discovery and development pathway of tirzepatide, its clinical trial progression, major therapeutic applications, and its early commercial impact. Evidence from early clinical studies and large Phase 3 programs demonstrated clinically meaningful improvements in glycemic control and body weight, supporting regulatory approval and rapid adoption. The product has also become commercially significant for Eli Lilly, with 2025 worldwide revenue of $22.965 billion for Mounjaro and $13.542 billion for Zepbound.

1. Introduction

Modern pharmaceutical development often begins with identification of a biological target followed by molecule design, laboratory evaluation, preclinical testing, and progressively larger human clinical trials. Tirzepatide provides a useful example of this pathway because its development combined established incretin biology with a novel dual-receptor pharmacological strategy. Tirzepatide, previously designated LY3298176, is a fatty-acid-modified peptide engineered for once-weekly subcutaneous administration. It acts at both the GIP and GLP-1 receptors. GIP and GLP-1 are incretin hormones involved in glucose-dependent insulin secretion and metabolic regulation. The development strategy was based on the hypothesis that simultaneous activity at both receptors could provide broader metabolic effects than selective GLP-1 receptor agonism alone. Early experimental work demonstrated activity at both receptors and supported further clinical evaluation. From a drug-development perspective, tirzepatide is especially relevant because its development demonstrates the importance of translating receptor pharmacology into a clinically practical dosage regimen. The molecule was not simply selected for receptor activity; its structure was also designed to support prolonged exposure and once-weekly administration. This combination of biological activity, pharmacokinetic optimization and clinical testing illustrates the multidisciplinary nature of modern pharmaceutical research.

2. Drug Discovery and Development Process

The discovery program for tirzepatide was built around incretin receptor biology. Researchers sought to determine whether adding GIP receptor activity to GLP-1 receptor activity could improve metabolic outcomes. The resulting molecule, LY3298176, was designed as a fatty-acid-modified peptide with dual GIP and GLP-1 receptor agonist activity and a pharmacokinetic profile suitable for once-weekly dosing. In vitro studies showed activation of both GIP and GLP-1 receptor signaling. In animal experiments, the compound produced glucose-dependent insulin secretion, improved glucose tolerance, and reduced food intake and body weight. These findings provided a preclinical proof of concept and justified progression to human studies. The first-in-human clinical program included Phase 1 studies using single-ascending and multiple-ascending doses in healthy participants, followed by a Phase 1b proof-of-concept study in people with T2DM. Across these studies, pharmacokinetic findings supported once-weekly administration. Dose-dependent reductions in glucose and body weight were observed, while the most frequently reported adverse effects were gastrointestinal, including nausea, vomiting and diarrhea. These results provided an initial balance of efficacy, tolerability and dosing information for later trials. The development then progressed into a broad Phase 3 program. The SURPASS trials evaluated tirzepatide in people with T2DM across different treatment settings, including monotherapy and use with background therapies. The SURMOUNT program subsequently evaluated tirzepatide for chronic weight management in people with obesity or overweight. This progression illustrates the conventional translational pathway from target selection and molecule optimization to preclinical evidence, early human safety and pharmacology studies, proof of concept, confirmatory efficacy trials and regulatory review.

3. Clinical Trial Phases

Clinical development of tirzepatide can be viewed through the standard sequence of Phase 1, Phase 2 and Phase 3 research. Phase 1: The early human program primarily assessed safety, tolerability, pharmacokinetics and dose-response relationships. The first-in-human program included single-ascending-dose and multiple-ascending-dose studies, followed by a Phase 1b study in participants with T2DM. The findings supported once-weekly subcutaneous dosing and demonstrated early effects on fasting glucose and body weight. Phase 2: Phase 2 studies explored dose selection and preliminary efficacy in patients with T2DM. A 26-week Phase 2b study evaluated several tirzepatide doses against placebo and dulaglutide. Results showed improvements in glycemic control and body weight and helped establish the dose range that was subsequently evaluated in Phase 3. Phase 3: The SURPASS program provided extensive evidence for diabetes treatment. In SURPASS-1, a 40-week randomized, double-blind, placebo-controlled Phase 3 trial, tirzepatide was studied as monotherapy in adults with T2DM. The trial demonstrated substantial reductions in HbA1c and body weight compared with placebo. Other SURPASS trials examined patients receiving background therapies and comparisons with established treatments. The SURMOUNT program extended development into obesity. SURMOUNT-1 was a 72-week Phase 3 randomized trial involving 2,539 adults with obesity or overweight and a weight-related complication, excluding diabetes. Mean body-weight reductions were approximately 15.0%, 19.5% and 20.9% with 5 mg, 10 mg and 15 mg tirzepatide, respectively, compared with 3.1% with placebo. Gastrointestinal adverse events were the most common and were generally mild to moderate, particularly during dose escalation. Regulatory approval followed the accumulated evidence. FDA approved Mounjaro on 13 May 2022 for improving glycemic control in adults with T2DM as an adjunct to diet and exercise. On 8 November 2023, FDA approved Zepbound for chronic weight management in adults with obesity or overweight with at least one weight-related condition. The FDA's Mounjaro review notes that nine clinical trials involving 7,769 adults with T2DM contributed to the approval, with five trials used for efficacy evaluation and all nine contributing to safety assessment.

4. Therapeutic Applications

The first FDA-approved therapeutic application of tirzepatide was T2DM. Mounjaro is indicated as an adjunct to diet and exercise to improve glycemic control in adults with T2DM. The drug is administered subcutaneously once weekly, with dose escalation designed to improve tolerability. Tirzepatide subsequently gained an important second therapeutic role in chronic weight management. Zepbound is indicated, together with a reduced-calorie diet and increased physical activity, for adults with obesity or adults who are overweight and have at least one weight-related condition. The approval reflects evidence that tirzepatide can produce substantial and sustained weight reduction. Beyond the original indications, clinical research continues to investigate tirzepatide in obesity-related and cardiometabolic conditions. Long-term SURMOUNT-1 follow-up published in 2025 reported sustained weight reduction over 176 weeks and a markedly lower rate of progression to T2DM among participants with obesity and prediabetes compared with placebo. These findings strengthen the clinical importance of tirzepatide for metabolic disease, although individual indications should always be distinguished from areas that remain under investigation. From a pharmaceutical perspective, tirzepatide is also an example of how molecular design, formulation strategy and receptor pharmacology can work together. Its modified peptide structure supports prolonged exposure, while the injectable dosage form enables once-weekly administration. The product therefore demonstrates the translation of molecular pharmacology into a practical chronic-disease therapy.

5. Market Impact

Tirzepatide has had a major commercial impact in the metabolic-disease market. Its dual-receptor mechanism, strong clinical efficacy and approvals in both diabetes and chronic weight management have supported rapid demand. The commercial impact is particularly visible in Eli Lilly's reported product revenues. According to Eli Lilly's 2025 results, worldwide Mounjaro revenue reached approximately $22.965 billion, compared with $11.540 billion in 2024, representing a 99% increase. Zepbound generated approximately $13.542 billion in 2025 compared with $4.926 billion in 2024, a 175% increase. Lilly reported that strong demand was a major driver of the increases. Together, these figures show that tirzepatide had become one of the company's most commercially important product franchises. Demand has also created supply and manufacturing challenges. FDA documentation regarding tirzepatide shortages notes that Mounjaro was added to the FDA drug shortage list in December 2022 because of high demand. This illustrates an important market-development issue: successful clinical adoption can create pressure on manufacturing capacity, distribution and product availability. The broader market significance of tirzepatide extends beyond sales. The drug has intensified competition and innovation in obesity and metabolic medicine, encouraging continued development of incretin-based therapies and alternative delivery approaches. Its success also demonstrates the commercial value of treatments that address both glycemic control and body-weight management. Nevertheless, market impact should be assessed alongside safety, affordability, access, supply and long-term clinical outcomes rather than revenue alone.

6. Conclusion

Tirzepatide represents a significant example of modern pharmaceutical drug development. Its discovery was driven by the hypothesis that combined GIP and GLP-1 receptor activation could improve metabolic outcomes. Preclinical studies supported the biological concept, early clinical studies established safety, pharmacokinetics and proof of concept, and extensive Phase 3 programs confirmed efficacy across T2DM and obesity populations. FDA approval of Mounjaro in 2022 and Zepbound in 2023 translated this evidence into clinical practice. The available evidence demonstrates substantial effects on glycemic control and body weight, while gastrointestinal adverse effects remain among the most common tolerability considerations. Commercially, the rapid growth of Mounjaro and Zepbound revenues and the documented supply pressure associated with high demand illustrate the strong market impact of tirzepatide. Overall, tirzepatide demonstrates how target-based discovery, rational molecular design, phased clinical evaluation and regulatory science can culminate in a high-impact pharmaceutical therapy. Continued post-marketing evidence and ongoing clinical research will be important for defining its long-term benefits, risks, access and role within the evolving treatment landscape.

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